

AME 34351 - Check-in 2 - Group 5

A. Team Members and Roles

Michael Doyle - Visualization and Modeling Lead
John Carlson - Deployment/Communication Lead
Peter Dacunto - Data Processing Lead
Sophia Bracken - Problem/Domain Lead

B. Link to Google Colab Notebook

https://drive.google.com/file/d/15Ql8WpGE4_Aoqux9pJI6yTcxkGlKtCTS/view?usp=sharing

1. Updated Problem Statement

The goal with this project is to first model the energy consumption, heating load, and/or cooling load of residential buildings using a variety of different architectural, geometric, and structural inputs (choices made during the buildings' early design stages). Then, the team will create a tool for building designers to input their own building characteristics and predict the energy consumption of the resulting building. The primary aim of the system is to help architects, engineers, and designers who want to design new and energy-efficient buildings minimize projected energy consumption for their designs.

The focus of the project has shifted from a commercial building energy analysis to a residential building energy analysis; in order to clearly define that focus, the team will utilize the NREL ResStock dataset, which identifies the characteristics of various residential buildings across the United States and compares them with recorded energy consumption values. The dataset is simulated, but it offers a streamlined collection of points, enabling a more complete analysis with our intended ML model design.

2. Confirmed Data Source and Dataset Description

Our confirmed data source (NREL ResStock — End-Use Load Profiles for U.S. Building Stock) can be found at <https://resstock.nrl.gov/>, a website which organizes the dataset into multiple packages based on U.S. states and the years in which data was recorded. The data is a statistically representative simulation (NOT measured utility bills). Each row represents one simulated dwelling archetype, weighted to reflect the real U.S. housing stock.

DATA STATISTICS:

204 input columns, 69 output columns, 0% missing

Target column name: out.site_energy.total.energy_consumption.kwh

Units: kWh — annual site energy consumption per dwelling

3. Documentation of Data Cleaning/Processing

1. Column Selection and Loading:

- Only relevant columns (target and selected features) are loaded from the parquet file, along with an optional weight column.
- Duplicate column names and those not present in `all\_columns` are removed.

2. Target Variable Cleaning:

- Rows with missing or non-numeric values in the `TARGET\_COL` are removed.
- The `TARGET\_COL` is converted to a numeric data type.
- Strip the 'F' suffix from heating/cooling setpoint strings (e.g., '68F' → 68) and cast those values to integer values.

3. Floor Area Parsing and Feature Engineering:

- A `parse\_floor\_area\_range` function is used to standardize the `FLOOR\_AREA\_COL` values, handling different formats like ranges ('1000-1499'), inequalities ('>4000', '<500'), and values with a '+' sign ('4000+'). For categorical plotting, these are returned as strings.
- For numerical calculations (specifically for `energy\_per\_floor\_area`), a numerical conversion is applied:
 - Ranges like '1000-1499' are converted to their midpoint (e.g., 1249.5).
 - Values like '>4000' are converted to 4000.
 - Values like '<500' are converted to their midpoint (e.g., 250).
 - Values like '4000+' are converted to 4000.
- A new feature, `energy\_per\_floor\_area`, is calculated by dividing the `TARGET\_COL` by the numerically converted floor area.

4. Meaningful Data Visualizations

Visualization I: No. of simulated dwelling models vs. annual site energy consumption

Location/Scope: State of Indiana

Description: This histogram reveals common trends among total energy consumption across Indiana, depicting the most common total annual energy consumption, based on the number of simulated dwellings with that energy consumption. Figure 1 depicts that the vast majority of buildings fall within the 15000 kWh to 30000 kWh range, indicating a non-categorical spread of total energy consumption. Interestingly enough, there is a dip of 200 or so buildings in the 18000 kWh to 22000 kWh range, which may require further analysis to determine a meaningful correlation.

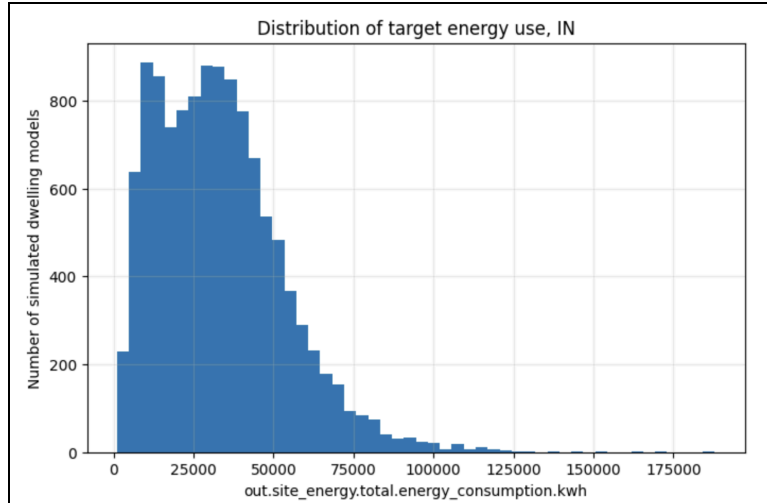


Figure 1. Visualization of Simulated Models vs. Annual Energy Consumption

Visualization II: Mean total energy consumption vs. geometric floor area

Location/Scope: National (entire United States)

Description: This histogram depicts mean total energy consumption per site as a variation of geometric floor area, indicating how energy consumption varies by square footage. The prevailing identifiable trend is that while larger and larger homes do require more energy, each successive step between floor area comes with a smaller increase in mean energy consumption, indicating a non-linear relationship. For instance, the jump in energy consumption from the 500-749 ft^2 bin to the 750-999 ft^2 bin is much more pronounced than the jump from 2500-2999 ft^2 to 3000-3999 ft^2 .

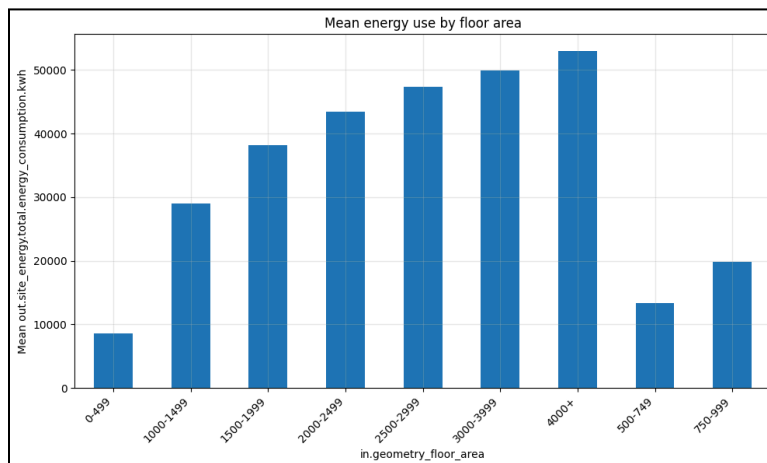


Figure 2. Visualization of Mean Energy Consumption vs. Geometric Floor Area

5. Data Challenge + Plan to Address

One challenge present in the data is high dimensionality. This simulated data set is very large, covering all 50 states. The data set has a total of 289 columns, a very large number. Currently, the notebook addresses this problem by filtering the data to remove columns with large amounts of missing data and columns with the same value for every row. Then, the model takes the first 25 columns, which are already ordered by importance, and uses them for the machine learning process. This process can be improved or changed by our group by using either SVD or t-SNE to address the high dimensionality of the data. t-SNE would be ideal for this data set because it has the goal of reducing high dimensional data and allowing for visualization. SVD could be used, but it is more focused on finding trends instead of reducing high dimensional data.

6. Updated Machine Learning Task Definition

The goal of the project is to develop a machine learning model that predicts a residential building's total annual energy consumption using building geometry, structural characteristics, and heating/cooling settings.

The input features describe the characteristics that influence building energy use. These include building geometry and structure, such as floor area, building type, number of stories, attic type, foundation type, unit location, and number of residential units which help characterize size, shape, and physical layout of the building. The machine learning model also includes building age characteristics such as construction period, which shows the differences that arise in building materials, insulation standards, and energy efficiency across periods of time. Heating and cooling characteristics are also included such as heating fuel type. All of these features provide information about the building's design and energy related factors that influence total annual energy consumption.

This machine learning task is a supervised regression problem where the model predicts a continuous numerical value for each energy consumption.

7. Plan for Data Train/Test Split

Randomly splitting the data with a fixed seed value (random_state=34351, matching the starter notebook) ensures full reproducibility across group (state) members. The test set is strictly held out, and touched only once to report final RMSE and R² in the final submission. All hyperparameter choices (Ridge α , RF max_depth, etc.) are made using the validation set only. We will use a 70/30 train test split. The random split will divide the data into these two sized groups, and the fixed seed will allow for reproducibility.